Excel — Top 25 interview questions

1. What are the differences between relative, absolute, and mixed cell references in Excel?

Answer:
In Excel, cell references define how formulas behave when copied to other cells.

- **Relative cell reference (A1):** Adjusts automatically when copied. It is used when the calculation logic should move with the data.
- **Absolute cell reference (A1):** Remains constant regardless of where the formula is copied. It is commonly used for fixed values such as tax rates or constants.
- **Mixed cell reference ($A1 or A$1):** Locks either the row or column while allowing the other to change. This is useful in matrix calculations and structured tables.

2. Explain the difference between COUNT, COUNTA, COUNTIF, and COUNTIFS functions.

Answer:

- **COUNT** counts only numeric values.
- **COUNTA** counts all non-empty cells, including text and dates.
- **COUNTIF** counts cells that satisfy a single condition.
- **COUNTIFS** counts cells that satisfy multiple conditions across one or more ranges.

These functions are commonly used in reporting and data validation.

3. How does VLOOKUP differ from INDEX + MATCH? Explain with advantages and disadvantages.

Answer:
VLOOKUP searches for a value in the first column of a table and returns a corresponding value from the right. It is easy to use but inflexible, as it cannot look left and breaks if columns are rearranged.

INDEX + MATCH is more flexible, allows lookup in any direction, and performs better on large datasets. Due to its robustness, INDEX + MATCH is preferred in professional environments.

4. Describe how XLOOKUP works. How can it be replaced if unavailable?

Answer:
XLOOKUP searches a range for a value and returns a matching result from another range. It supports left and right lookups, handles missing values gracefully, and replaces multiple legacy lookup functions.

If XLOOKUP is unavailable, its functionality can be replicated using INDEX + MATCH.

5. Explain how PivotTables are created and used. What are calculated fields?

Answer:
A PivotTable is created by selecting a dataset and choosing Insert → PivotTable. Fields are dragged into rows, columns, values, and filters to summarize data.

Calculated fields allow users to perform custom calculations within a PivotTable, such as calculating profit using sales and cost fields.

6. How do you remove duplicate records in Excel, and what are the data integrity concerns?

Answer:
Duplicates can be removed using the Remove Duplicates tool, Advanced Filters, or Power Query. Removing duplicates permanently deletes data, which can affect data integrity.

Best practice is to create a backup or use Power Query to preserve original data.

7. Explain conditional formatting and its practical use with formulas.

Answer:
Conditional formatting automatically formats cells based on rules. It is often used to highlight exceptions such as low sales or overdue tasks.

Formulas can be combined with conditional formatting to apply advanced logic, such as highlighting values below a dynamic target.

8. What is Data Validation, and how is it used to restrict input?

Answer:
Data Validation controls the type of data entered into cells. It can restrict entries to numeric ranges, dates, lists, or custom formulas.

A common use is drop-down lists to standardize inputs and reduce data entry errors.

9. Write and explain a formula to extract the nth word from a text string.

Answer:
Excel can extract the nth word using a combination of SUBSTITUTE, MID, and TRIM functions. This method replaces spaces with large gaps and extracts the required portion.

This technique is useful in text processing and data cleaning.

10. Explain common text functions with an example.

Answer:
Excel text functions manipulate and clean text data:

- LEFT and RIGHT extract characters from either end.
- MID extracts text from the middle.
- TRIM removes extra spaces.
- LEN counts characters.
- FIND identifies the position of text.

These functions are commonly used in data preparation tasks.

11. How do you handle messy date formats imported from external systems?

Answer:
Messy dates are cleaned using Text to Columns, DATEVALUE, or Power Query. Regional format issues such as MM/DD and DD/MM are resolved by explicitly defining date formats.

Power Query is preferred for handling large or recurring imports.

12. Explain SUMIFS and AVERAGEIFS with use cases.

Answer:
SUMIFS and AVERAGEIFS calculate totals or averages based on multiple conditions. They are widely used in financial analysis, sales reporting, and performance tracking.

13. What are array formulas and dynamic arrays? When should they be used?

Answer:
Array formulas perform calculations on multiple values simultaneously. Dynamic array functions such as FILTER and UNIQUE automatically spill results into adjacent cells.

They are used in modern Excel dashboards and dynamic reports.

14. Describe the process of building a dynamic dashboard in Excel.

Answer:
A dynamic dashboard combines PivotTables, charts, slicers, and formulas to present key metrics. Best practices include a clean layout, minimal colors, and business-focused KPIs.

Dashboards should provide quick insights rather than raw data.

15. Explain macros and differentiate between VBA and Power Query automation.

Answer:
Macros automate repetitive tasks. VBA is used for interface-level automation, while Power Query is used for data transformation.

Power Query is preferred for ETL tasks, whereas VBA is used for user-driven actions.

16. What is Power Query, and how does it differ from Excel formulas?

Answer:
Power Query is an ETL tool that cleans and transforms data before it is loaded into Excel.

Unlike formulas, it does not recalculate with every change and is more efficient for large datasets.

17. Explain Power Pivot and the Excel Data Model.

Answer:
Power Pivot allows users to create relationships between multiple tables. Calculated columns operate row-wise, while measures aggregate data dynamically using DAX.

Measures are preferred for performance and scalability.

18. What are common performance issues in large Excel files, and how can they be resolved?

Answer:
Performance issues arise from excessive formulas, volatile functions, and large formatting ranges. These can be mitigated by using tables, reducing formulas, and leveraging Power Pivot.

19. How do you create a consolidated report from multiple worksheets?

Answer:
Consolidation can be done using Power Query, VBA, or Excel's Consolidate tool. Power Query is the most reliable and scalable method.

20. Explain workbook and worksheet protection and their limitations.

Answer:
Excel allows protection of worksheets and workbooks using passwords. While useful for preventing accidental edits, these protections are not suitable for high-security requirements.

21. Explain nested IF formulas and better alternatives.

Answer:
Nested IF formulas handle multiple conditions but are difficult to maintain. Alternatives such as IFS or lookup functions improve readability and scalability.

22. What are Goal Seek and Solver? Provide business use cases.

Answer:
Goal Seek finds the input needed to reach a target output. Solver optimizes outcomes under constraints. They are used in financial modeling and optimization problems.

23. Explain how formula auditing tools help in Excel.

Answer:
Formula auditing tools trace dependencies, identify errors, and evaluate calculations step by step. They are essential for debugging complex spreadsheets.

24. How would you prepare a 1-million-row CSV file for analysis in Excel?

Answer:
Large datasets should be handled using Power Query and the Data Model. Formulas should be minimized, and analysis should be done using PivotTables

25. Explain how to reconcile two lists using fuzzy matching.

Answer:
Fuzzy matching identifies approximate matches between lists using Power Query's Fuzzy Merge. It is commonly used in vendor and customer reconciliation.